

CS173: Discrete Structures

Proving induction

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Theorem (Weak Induction). For any wff φ with one free variable, the following equivalence holds:

$$(\forall n \in \mathbb{N})(\varphi(n)) \iff \varphi(0) \wedge (\forall k \in \mathbb{N})(\varphi(k) \Rightarrow \varphi(S(k)))$$

Theorem.

$$\forall M((M \subseteq \mathbb{N} \wedge M \neq \emptyset) \Rightarrow \exists m(m \in M \wedge (\forall n \in M)(m \leq n)))$$

Proof. $(\Leftarrow)$ Assume $\varphi(0)$ and $(\forall k \in \mathbb{N})(\varphi(k) \Rightarrow \varphi(S(k)))$.

Let $n \in \mathbb{N}$. Towards a contradiction, assume $\neg\varphi(n)$.

Consider $M := \{m \mid m \in \mathbb{N} \wedge \neg\varphi(m)\}$. Observe $M \subseteq \mathbb{N}$.

Observe $n \in M$ (because $n \in \mathbb{N}$ and $\neg\varphi(n)$). Therefore $M \neq \emptyset$.

So, there exists $m \in M$ such that $(\forall x \in M)(m \leq x)$.

Case 1: Suppose $m = 0$. Recall $\varphi(0)$, so $\varphi(m)$. However, $\neg\varphi(m)$ because $m \in M$. $\Rightarrow \Leftarrow$

Case 2: Suppose there exists $w \in \mathbb{N}$ such that $S(w) = m$. This means $w \cup \{w\} = m$. So $w \in m$. So $w < m$.

Case 2.1: Suppose $\neg\varphi(w)$. Then $w \in M$. Since $(\forall a \in M)(m \leq a)$, we have $m \leq w$.

By definition, $m \in w$ or $m = w$. However, $m \neq w$.

So $m \in w$. However, $\forall x \forall y (x \in y \Rightarrow y \notin x)$. $\Rightarrow \Leftarrow$

Case 2.2: Suppose $\varphi(w)$. Then, by assumption, we know $\varphi(w) \Rightarrow \varphi(S(w))$.

So $\varphi(S(w))$. So $\varphi(m)$. However, $\neg\varphi(m)$. $\Rightarrow \Leftarrow$

$(\Rightarrow)$ Assume $(\forall n \in \mathbb{N})(\varphi(n))$.

Since $0 \in \mathbb{N}$, we know $\varphi(0)$.

Let $k \in \mathbb{N}$. Assume $\varphi(k)$.

Since $k \in \mathbb{N}$, we know $S(k) \in \mathbb{N}$. Therefore $\varphi(S(k))$.

□

Theorem.

$$(\forall n \in \mathbb{N})(1 + n = S(n))$$

Proof. By weak induction on n .

Base Case: Observe

$$\begin{aligned} 1 + 0 &= 1 && \text{by definition of addition} \\ &= S(0) && \text{by definition of } 1 \end{aligned}$$

Inductive Step: Let $k \in \mathbb{N}$.

We need to show (NTS): $1 + S(k) = S(S(k))$, assuming the inductive hypothesis $1 + k = S(k)$.

Assume $1 + k = S(k)$. Observing:

$$\begin{aligned} 1 + S(k) &= S(1 + k) && \text{by definition of addition} \\ &= S(S(k)) && \text{by the inductive hypothesis} \end{aligned}$$

Therefore, $(\forall n \in \mathbb{N})(1 + n = S(n))$.

□